SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE	DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

Program Name: B. Tech	**Assignment Type: Lab**	**AcademicYear:**2025-2026
Course Coordinator Name	Dr. Rishabh Mittal	
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Course Code	23CS002PC304	**CourseTitle**	AI Assisted Coding
Year/Sem	III/II	**Regulation**	R23
Date and Day of Assignment	Week12 - Tuesday	**Time(s)**	
Duration	2 Hours	**Applicable to Batches**	23CSBTB01 To 23CSBTB52

Assignment Number:24.2(Present assignment number)/**24**(Total number of assignments)

Q.No.	Question	*Expected Time to complete*
1	**Lab 24 – Replit for Mobile Applications** **Lab Objectives:** • Explore **Replit** as a cloud-based IDE for building mobile applications. • Learn how to leverage AI-assisted coding inside Replit for faster development. • Understand how to design, test, and deploy simple mobile apps. • Gain experience in collaborative app development directly from the browser.	Week 12 - Tuesday
	Task 1 – Environment Setup	

	• Create a free **Replit account**. • Explore the mobile app development templates (React Native / Flutter / HTML5 hybrid). • Install required packages or dependencies in the Replit environment.	

	Task 2 – Basic Mobile App
	• Build a simple **"Hello App"** that displays:
	  ○ App Title
	  ○ User's Name (input field + button)
	  ○ Greeting message dynamically updated.
	• Use AI assistance to generate boilerplate code and explain structure.	

	Task 3 – Adding Features with AI Assistance
	• Enhance the app with:
	  ○ **Navigation bar** (Home, About, Contact).
	  ○ **Form handling** (e.g., user registration page).
	  ○ **Basic styling** with CSS or React Native styles.
	• Ask AI in Replit to **add inline documentation** for each function.	

	Task 4 – Testing on Mobile • Run the app in Replit's mobile preview mode. • Test responsiveness on different screen sizes. • Fix UI alignment issues with AI suggestions.	
	Task 5 – Deployment • Deploy the app using **Replit hosting** or export it to a mobile framework (like Expo for React Native). • Share the deployment link with peers for testing.	
	Expected Outcomes: • Students will understand how to use **Replit for mobile app development**. • Ability to build a **basic functional mobile app** with UI, navigation, and user interaction. • Experience in **AI-assisted coding, debugging, and documentation** for mobile applications.	

Task 1 :

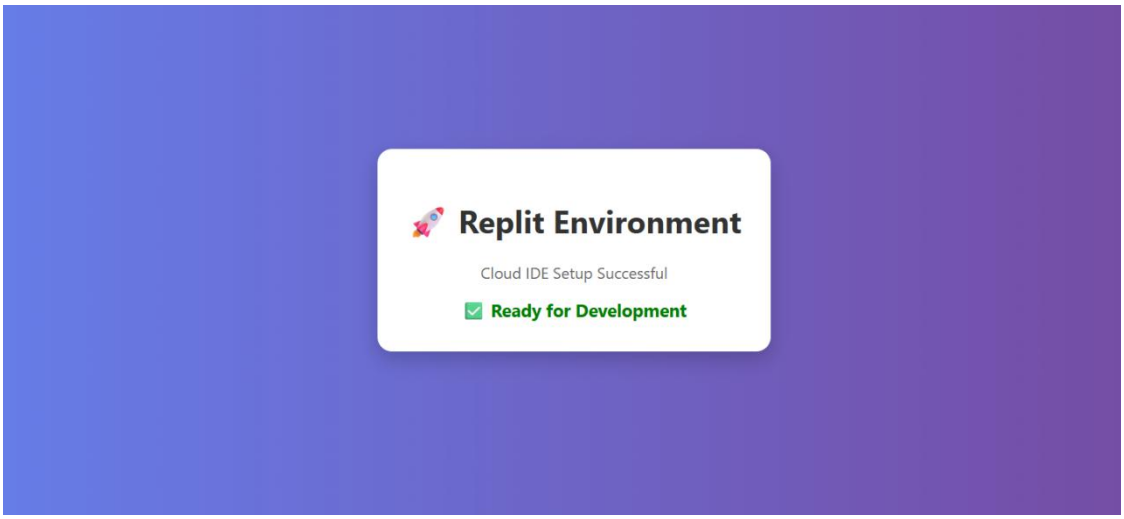
Prompt:

Set up a Replit environment for mobile application development using HTML5 or React-based templates. Install required dependencies and prepare the workspace for AI-assisted coding.

Code:

```html
<!DOCTYPE html>
<html>
<head>
    <title>Replit Setup</title>

    <style>
        body {
            margin: 0;
            font-family: Arial, sans-serif;
            background: linear-gradient(to right, #667eea, #764ba2);
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .card {
            background: white;
            padding: 30px;
            border-radius: 15px;
            text-align: center;
            box-shadow: 0 8px 20px rgba(0,0,0,0.2);
        }

        h1 {
            color: #333;
        }

        p {
            color: #666;
        }

        .status {
            margin-top: 15px;
            font-size: 18px;
            color: green;
            font-weight: bold;
        }
    </style>
</head>

<body>

<div class="card">
    <h1>🚀 Replit Environment</h1>
    <p>Cloud IDE Setup Successful</p>
    <div class="status">✅ Ready for Development</div>
</div>

</body>
</html>
```

Sample Input/Output:



Code Explanation:

1. This task sets up the development environment using Replit.
2. Replit is a cloud-based IDE that runs in the browser.
3. It supports mobile app development using web technologies.
4. Templates like HTML, React, or Flutter can be selected.
5. The basic test confirms the environment is working.
6. No local installation is required.
7. AI assistance in Replit helps generate code quickly.
8. This setup is required for further tasks.

Task 2 : Basic Mobile App

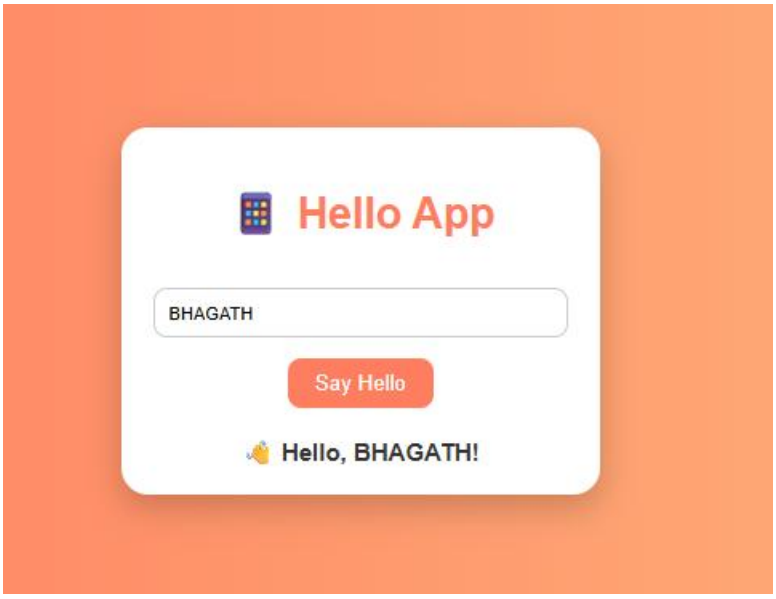
Prompt:

Build a simple mobile-style web app that displays an app title, takes user input, and shows a dynamic greeting message. Use AI assistance to generate colorful UI and interactive behavior.

Code:

```html
<!DOCTYPE html>
<html>
<head>
    <title>Hello App</title>

    <style>
        body {
            margin: 0;
            font-family: Arial;
            background: linear-gradient(to right, #ff7e5f, #feb47b);
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }

        .app {
            background: white;
            width: 320px;
            padding: 20px;
            border-radius: 20px;
            text-align: center;
            box-shadow: 0 10px 25px rgba(0,0,0,0.2);
        }

        h1 {
            color: #ff7e5f;
        }

        input {
            width: 90%;
            padding: 10px;
            margin-top: 15px;
            border-radius: 10px;
            border: 1px solid #ccc;
            outline: none;
        }

        button {
            margin-top: 15px;
            padding: 10px 20px;
            background: #ff7e5f;
            color: white;
            border: none;
            border-radius: 10px;
            font-size: 16px;
            cursor: pointer;
        }

        button:hover {
            background: #ff5733;
        }

        #message {
            margin-top: 20px;
            font-size: 18px;
            color: #333;
            font-weight: bold;
        }
    </style>
</head>

<body>

<div class="app">
    <h1>📱 Hello App</h1>

    <input type="text" id="name" placeholder="Enter your name">

    <button onclick="showMessage()">Say Hello</button>

    <div id="message"></div>
</div>

<script>
function showMessage() {
    let name = document.getElementById("name").value;

    if(name === "") {
        document.getElementById("message").innerHTML = "⚠️ Please enter your name!";
    } else {
        document.getElementById("message").innerHTML = "👋 Hello, " + name + "!";
    }
}
</script>

</body>
</html>
```

Sample Input/Output:



Code Explanation:

1. This task builds a simple interactive mobile-style app.
2. The UI is designed using HTML and CSS with a gradient background.
3. A centered card layout improves visual appearance.
4. The input field takes the user's name.
5. A button triggers a JavaScript function.
6. The function dynamically updates the greeting message.
7. Input validation ensures proper user interaction.
8. This demonstrates basic UI + logic integration.

Task 3 : Adding Features with AI Assistance

Prompt:

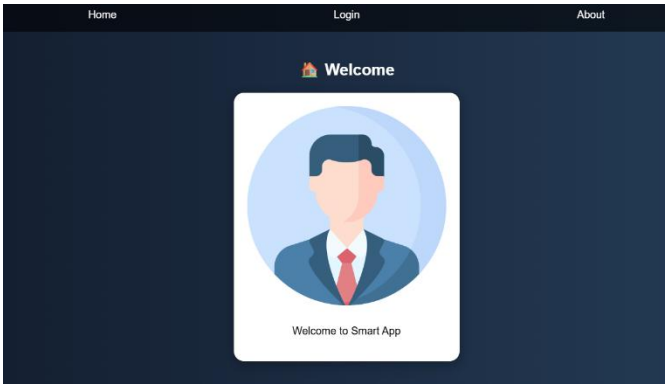
Enhance the mobile app by adding navigation, form handling, and improved styling using AI assistance. Include inline documentation for each function to improve readability.

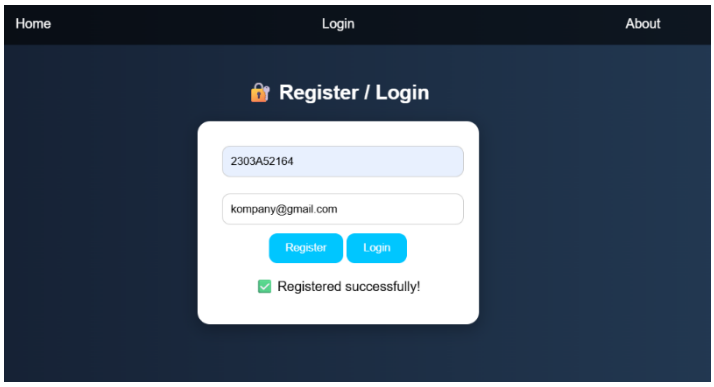
Code:

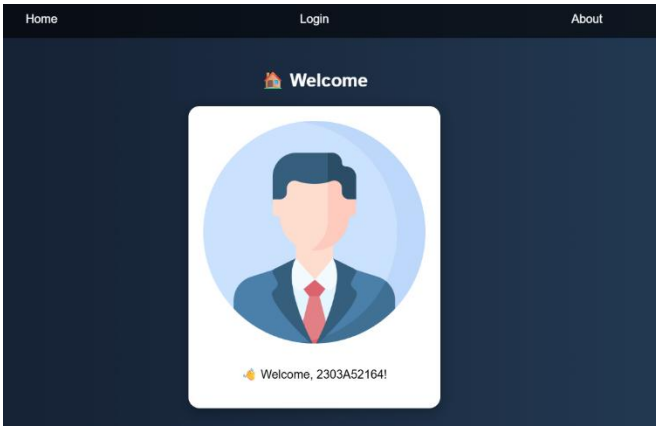
```html
<!DOCTYPE html>
<html>
<head>
  <title>Smart Mobile App</title>

  <style>
      body {
          margin: 0;
          font-family: Arial;
          background: linear-gradient(to right, #141e30, #243b55);
          color: white;
      }

      /* Navbar */
      .navbar {
          display: flex;
          justify-content: space-around;
          background: rgba(0,0,0,0.6);
          padding: 15px;
      }

      .navbar button {
          background: none;
          border: none;
          color: white;
          font-size: 16px;
          cursor: pointer;
      }

      .navbar button:hover {
          color: #00c6ff;
      }

      /* Page */
      .page {
          display: none;
          text-align: center;
          padding: 20px;
      }

      .active {
          display: block;
      }

      /* Card */
      .card {
          background: white;
          color: black;
          padding: 20px;
          margin: auto;
          width: 300px;
          border-radius: 15px;
          box-shadow: 0 5px 20px rgba(0,0,0,0.3);
      }

      input {
          width: 90%;
          padding: 10px;
          margin: 10px;
          border-radius: 10px;
          border: 1px solid #ccc;
      }

      button.submit {
          padding: 10px 20px;
          background: #00c6ff;
          border: none;
          border-radius: 10px;
          color: white;
          cursor: pointer;
      }

      button.submit:hover {
          background: #0072ff;
      }

      img {
          width: 100%;
          border-radius: 10px;
          margin-bottom: 10px;
      }
  </style>
</head>

<body>
```

```html
<!-- Navbar -->
<div class="navbar">
    <button onclick="showPage('home')">Home</button>
    <button onclick="showPage('login')">Login</button>
    <button onclick="showPage('about')">About</button>
</div>

<!-- Home -->
<div id="home" class="page active">
    <h2>🏠 Welcome</h2>
    <div class="card">
        <img src="https://cdn-icons-png.flaticon.com/512/3135/3135715.png">
        <p id="welcomeMsg">Welcome to Smart App</p>
    </div>
</div>

<!-- Registration -->
<div id="login" class="page">
    <h2>🔐 Register / Login</h2>
    <div class="card">

        <input type="text" id="username" placeholder="Enter Name">
        <input type="email" id="email" placeholder="Enter Email">

        <button class="submit" onclick="registerUser()">Register</button>
        <button class="submit" onclick="loginUser()">Login</button>

        <p id="output"></p>
    </div>
</div>

<!-- About -->
<div id="about" class="page">
    <h2>ℹ️ About</h2>
    <div class="card">
        <img src="https://cdn-icons-png.flaticon.com/512/4712/4712027.png">
        <p>This app is built using AI-assisted coding in Replit.</p>
    </div>
</div>

<script>

    // Store user data temporarily
    let storedUser = "";

    // Function to switch pages
    function showPage(pageId) {
        let pages = document.querySelectorAll(".page");
        pages.forEach(p => p.classList.remove("active"));
        document.getElementById(pageId).classList.add("active");
    }

    // Function to register user
    function registerUser() {
        let name = document.getElementById("username").value;
        let email = document.getElementById("email").value;

        if(name === "" || email === "") {
            document.getElementById("output").innerHTML = "⚠️ Fill all fields!";
            return;
        }

        storedUser = name; // Save name

        document.getElementById("output").innerHTML =
            "✅ Registered successfully!";
    }

    // Function to login user
    function loginUser() {
        let name = document.getElementById("username").value;

        if(name === storedUser && name !== "") {
            document.getElementById("welcomeMsg").innerHTML =
                "👋 Welcome, " + name + "!";
            showPage("home");
        } else {
            document.getElementById("output").innerHTML =
                "❌ User not found. Register first!";
        }
    }

</script>

</body>
</html>
```

Sample Input/Output:







Code Explanation:

1. This task enhances the mobile app with additional features.
2. A navigation bar allows switching between pages.
3. JavaScript is used to control page visibility dynamically.
4. A registration form collects user details.
5. Input validation ensures correct data entry.
6. CSS improves UI with gradients and responsive layout.
7. Inline comments explain each function clearly.
8. This demonstrates multi-page app behavior in a single file.

Task 4 :

Prompt:

Test the mobile app in Replit's mobile preview mode and ensure it is responsive across different screen sizes. Fix UI alignment issues using AI suggestions and responsive design techniques.

Code:

```
<style>

/* Make everything responsive */
body {
    margin: 0;
    font-family: Arial;
    background: linear-gradient(to right, #141e30, #243b55);
    color: white;
}

/* Navbar */
.navbar {
    display: flex;
    justify-content: space-around;
    flex-wrap: wrap;
    background: rgba(0,0,0,0.6);
    padding: 10px;
}

.navbar button {
    background: none;
    border: none;
    color: white;
    font-size: 14px;
    margin: 5px;
}

/* Pages */
.page {
    display: none;
    padding: 15px;
}

.active {
    display: block;
}

/* Card */
.card {
    background: white;
    color: black;
    padding: 15px;
    margin: auto;
    width: 90%;        /* responsive width */
    max-width: 350px;  /* limit for large screens */
    border-radius: 15px;
    box-shadow: 0 5px 20px rgba(0,0,0,0.3);
}

/* Inputs */
input {
    width: 90%;
    padding: 10px;
    margin: 10px 0;
    border-radius: 10px;
    border: 1px solid #ccc;
}

/* Buttons */
button.submit {
    width: 90%;
    padding: 10px;
    margin: 5px 0;
    background: #00c6ff;
    border: none;
    border-radius: 10px;
    color: white;
}

/* Images */
img {
    width: 100%;
    height: auto;
    border-radius: 10px;
}

/* Mobile Optimization */
@media (max-width: 500px) {
    h2 {
        font-size: 18px;
    }

    .navbar button {
        font-size: 12px;
    }
}
```

Sample Input/Output:

Code Explanation:

1. This task focuses on testing the application on mobile devices.
2. Replit provides a mobile preview for testing responsiveness.
3. The UI is checked across different screen sizes.
4. Issues like overflow and misalignment are identified.
5. CSS media queries are used to fix responsiveness.
6. Flexible layouts ensure proper display on all devices.
7. Images and buttons are resized dynamically.
8. This improves user experience across platforms.

Task 5 : Deployment

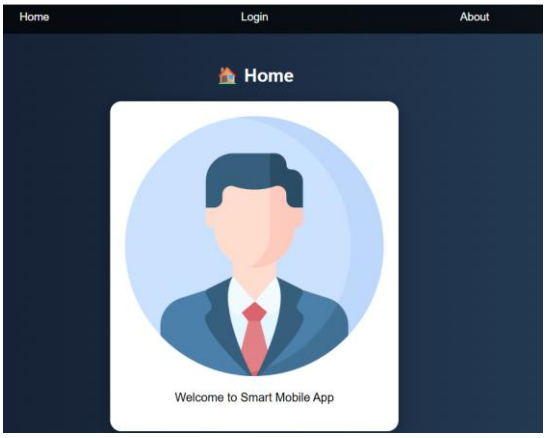
Prompt:

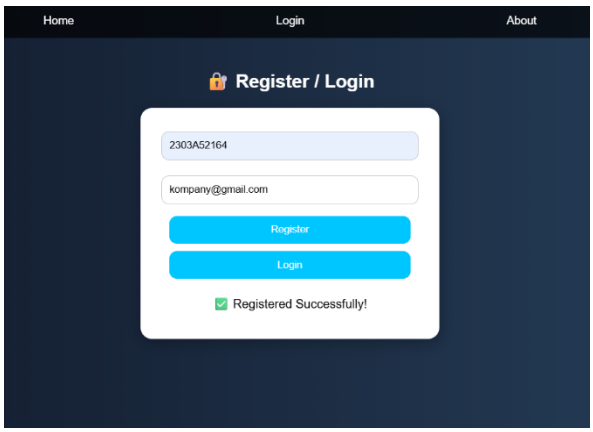
Deploy the developed mobile web application using Replit hosting and make it accessible via a public link. Test the deployed app and share the link for peer evaluation.

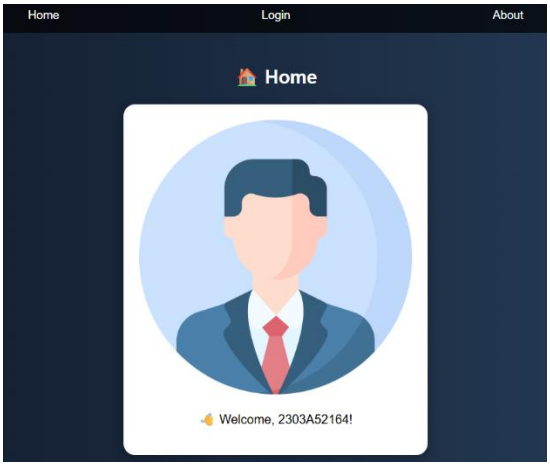
Code:

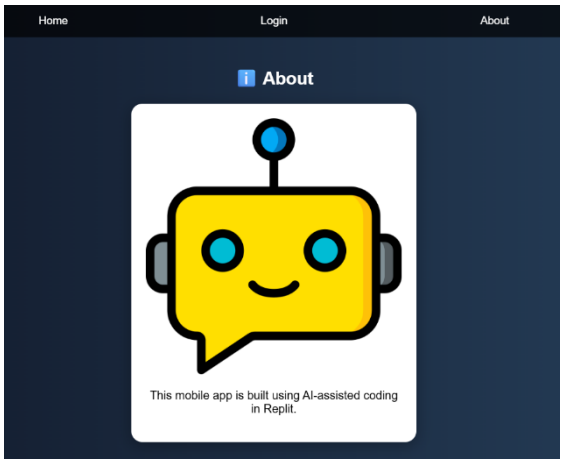
```html
<!DOCTYPE html>
<html>
<head>
    <title>Smart Mobile App</title>

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <style>
        body {
            margin: 0;
            font-family: Arial;
            background: linear-gradient(to right, #141e30, #243b55);
            color: white;
        }

        /* NAVBAR */
        .navbar {
            display: flex;
            justify-content: space-around;
            flex-wrap: wrap;
            background: rgba(0,0,0,0.7);
            padding: 12px;
        }

        .navbar button {
            background: none;
            border: none;
            color: white;
            font-size: 15px;
            cursor: pointer;
        }

        .navbar button:hover {
            color: #00c6ff;
        }

        /* PAGE */
        .page {
            display: none;
            padding: 20px;
            text-align: center;
        }

        .active {
            display: block;
        }

        /* CARD */
        .card {
            background: white;
            color: black;
            padding: 20px;
            margin: auto;
            width: 90%;
            max-width: 350px;
            border-radius: 15px;
            box-shadow: 0 5px 20px rgba(0,0,0,0.3);
        }

        /* INPUT */
        input {
            width: 90%;
            padding: 10px;
            margin: 10px 0;
            border-radius: 10px;
            border: 1px solid #ccc;
        }

        /* BUTTON */
        button.submit {
            width: 90%;
            padding: 10px;
            margin: 5px 0;
            background: #00c6ff;
            border: none;
            border-radius: 10px;
            color: white;
            cursor: pointer;
        }

        button.submit:hover {
```

Sample Input/Output:









Code Explanation:

1. This task focuses on deploying the developed application.
2. Replit provides free hosting for web applications.
3. The application is made accessible via a public URL.
4. Users can test the app in real-time using the link.
5. Deployment ensures the app works outside the development environment.
6. It allows sharing with peers for feedback.
7. AI assistance helped speed up development and debugging.
8. This completes the full mobile app development lifecycle.